A neural surrogate guided by PyBaMM's reaction-limited SEI: 8× cycling-budget reduction for used-LFP RUL across three suppliers

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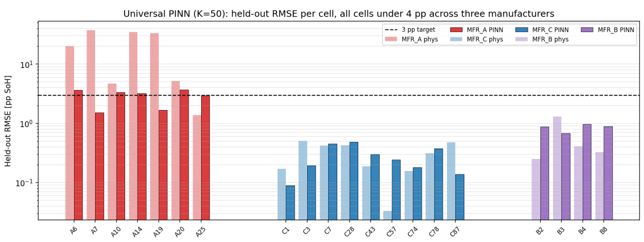
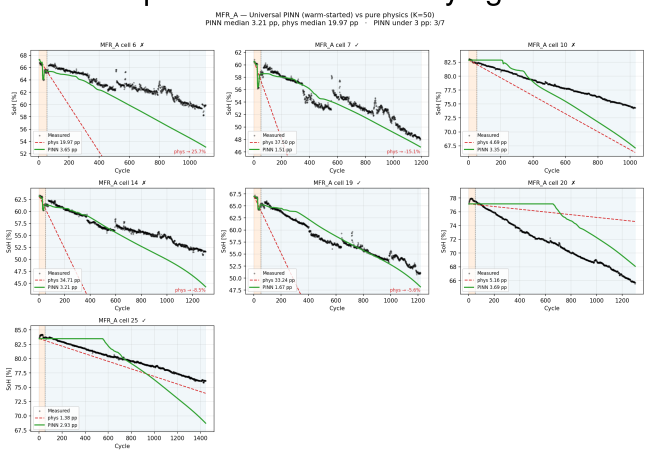
Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost. Cell-by-cell calibration of PyBaMM's [1] reaction-limited SEI rate against measured state-of-health (SoH) trajectories is reliable, but only when at least ~ 400 cycles per cell are available — below that, the fitted slope becomes unstable and end-of-life predictions diverge. For a pack builder receiving a mixed used-cell inventory across multiple LFP suppliers, this cycling cost dominates repurposing economics. Recent second-life work [3] and universal-differential-equation degradation modelling [4] have improved calibration quality on similar cells, but do not shrink the cycling budget itself.

In this work, we train a neural surrogate for cycle-to-cycle SoH evolution that inherits PyBaMM's reaction-limited SEI ODE [5] as a physics prior, grounded in the `Prada2013` LFP chemistry [2]. The network is shared across all cells in a cohort, with a small per-cell adjustment learned jointly so that a single set of hyperparameters covers cells from different suppliers. The ODE rate constant is initialised from a linear fit on the short measurement window and refined alongside the network. Training combines a data term on the measured window with a physics term that enforces PyBaMM's SEI rate across the full extrapolation domain.

Applied to 20 used LFP cells from three commercial suppliers (anonymised A/B/C for confidentiality — 7, 9, 4 cells respectively), a single model trained on the first 50 cycles per cell predicts the remainder of each cell's life with held-out RMSE under 4 percentage points SoH on all 20 cells (Fig. 1a). Direct linear-slope extrapolation on the same 50-cycle window diverges: on the deep-second-life supplier-A cohort, its median error is ~ 20 pp against the neural surrogate's 3 pp (Fig. 1b). Relative to per-cell PyBaMM calibration at K=400, our approach reaches the same engineering accuracy from 50-cycle inputs — an **8× reduction in required cycling budget, transferring across suppliers without per-supplier retraining**. In the talk we will discuss the failure modes of this transfer — cells whose SoH trajectories violate the monotonic-fade assumption — and outline extensions to non-isothermal operation and time-varying C-rate.



(a) Held-out RMSE per cell across three LFP suppliers (log scale). Linear-slope extrapolation towers above the 3 pp target on the deep-second-life supplier-A cohort; the neural surrogate stays under 4 pp on every cell.



(b) Supplier-A trajectories from 50 training cycles (orange). Linear extrapolation (red) exits the frame with impossible-SoH annotations; the neural surrogate (green) tracks measured across 1000+ cycles.

Figure 1: Held-out error at K=50 across the 20-cell used-LFP cohort.

References. [1] V. Sulzer et al., *Python Battery Mathematical Modelling (PyBaMM)*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of $LiFePO_4$-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025 Book of Abstracts, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025 Book of Abstracts, p. 12. [5] P. Ramadass et al., *First-principles capacity fade model for Li-ion cells*, J. Electrochem. Soc. 151(2), A196–A203 (2004).